LABCYCLE

submitted by
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II BCA

BCA2025 III SEMESTER
Python programming

LAB-CYCLE-1

1.Write a program to count
a.vowels,consonants,digits and special characters in a string given as input.
2.Write a program to find the largest and second largest elements in a list.
3.Write a program to count word frequency using a dictionary.
4.Write a program to sort dictionary by keys and the by values
5.Write a program using list comprehension to generate prime numbers.

AIM:1.Write a program to count
a.vowels,consonants,digits and special characters in a string given as input.

PROGRAM:

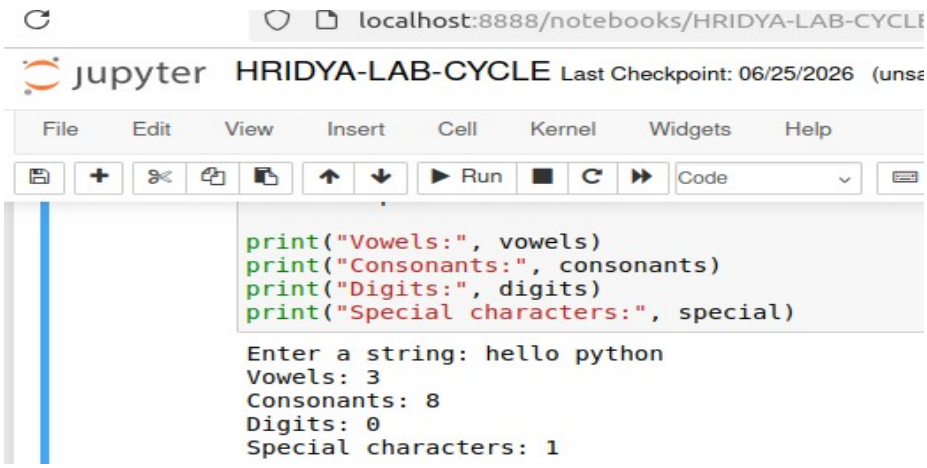
```
string = input("Enter a string: ")

vowels = 0
consonants = 0
digits = 0
special = 0

for ch in string:
    if ch.isalpha():
        if ch.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1
    elif ch.isdigit():
        digits += 1
    else:
        special += 1

print("Vowels:", vowels)
print("Consonants:", consonants)
print("Digits:", digits)
print("Special characters:", special)
```

OUTPUT:

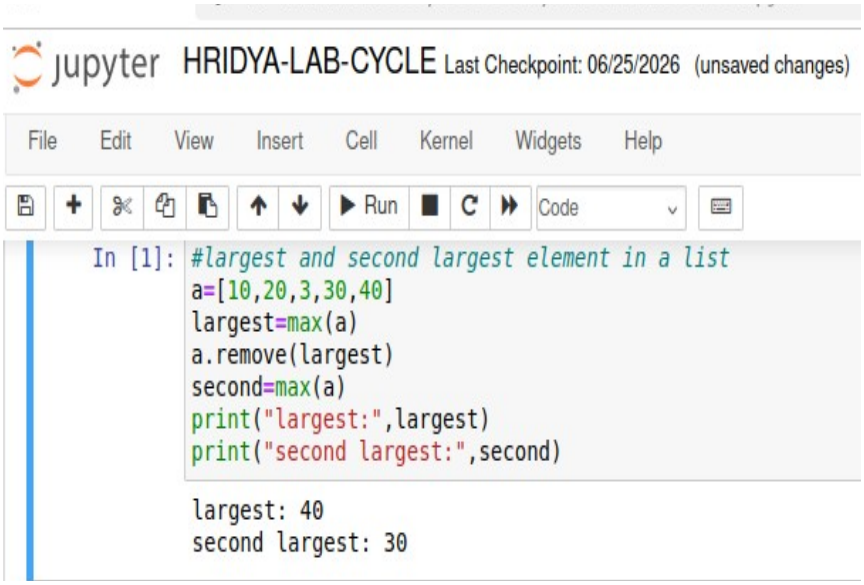


AIM:2.Write a program to find the largest and second largest elements in a list.

PROGRAM:

```
#largest and second largest element in a list
a=[10,20,3,30,40]
largest=max(a)
a.remove(largest)
second=max(a)
print("largest:",largest)
print("second largest:",second)
```

OUTPUT:



AIM:3.Write a program to count word frequency using a dictionary

PROGRAM:

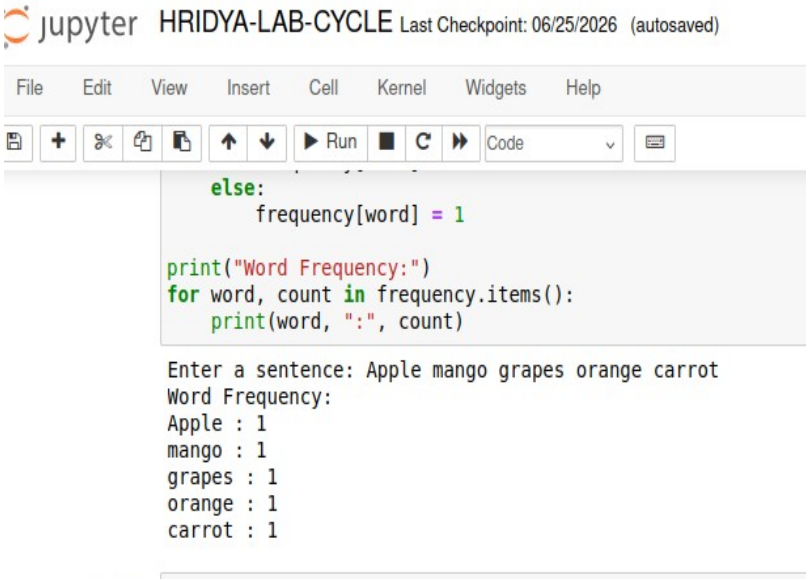
```
text = input("Enter a sentence: ")

words = text.split()
frequency = {}

for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1

print("Word Frequency:")
for word, count in frequency.items():
    print(word, ":", count)
```

OUTPUT:



AIM:4. Write a program to sort dictionary by keys and the by values.

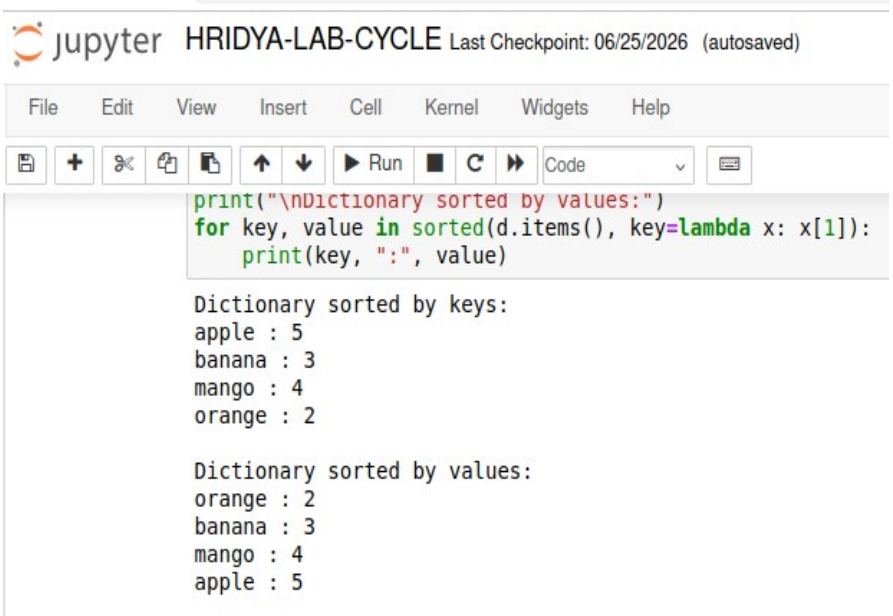
PROGRAM:

```
d = {"banana": 3, "apple": 5, "orange": 2, "mango": 4}

# Sort by keys
print("Dictionary sorted by keys:")
for key in sorted(d):
    print(key, ":", d[key])

# Sort by values
print("\nDictionary sorted by values:")
for key, value in sorted(d.items(), key=lambda x: x[1]):
    print(key, ":", value)
```

OUTPUT:



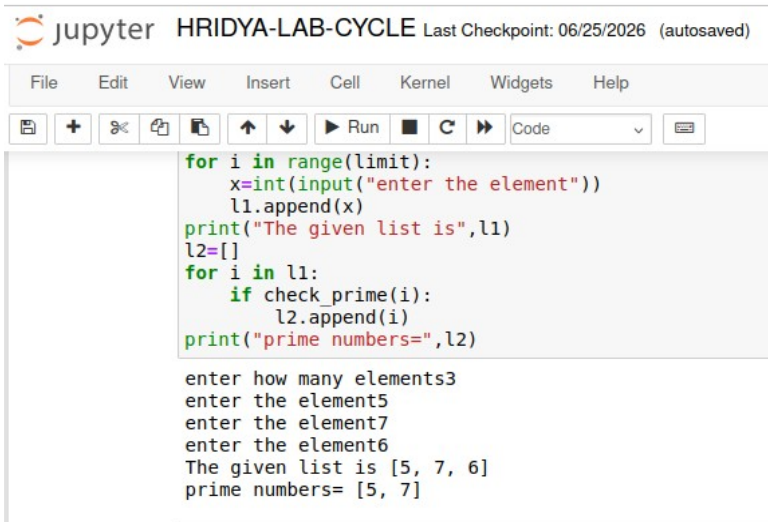
AIM:5.Write a program using list comprehension to generate prime numbers.

PROGRAM:

```
#WITHOUT USING LIST COMPREHENSION
def check_prime(num):
    if(num<=1):
        return False
    for i in range(2,(num//2)+1):
        if num%i==0:
            return False
    return True

#MAIN PROGRAM
l1=[]
limit=int(input("enter how many elements"))
for i in range(limit):
    x=int(input("enter the element"))
    l1.append(x)
print("The given list is",l1)
l2=[]
for i in l1:
    if check_prime(i):
        l2.append(i)
print("prime numbers=",l2)
```

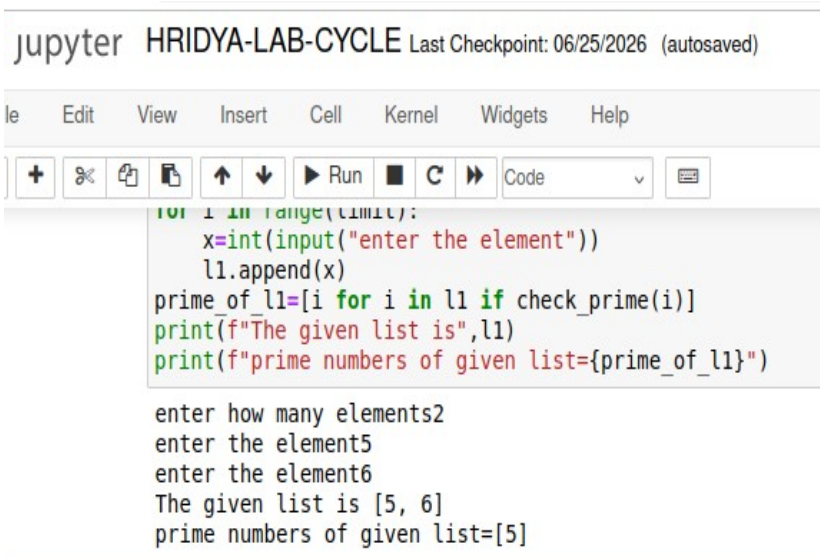
OUTPUT:



```
#USING LIST COMPREHENSION
def check_prime(num):
    if(num<=1):
        return False
    for i in range(2,(num//2)+1):
        if num%i==0:
            return False
    return True

#MAIN PROGRAM
l1=[]
limit=int(input("enter how many elements"))
for i in range(limit):
    x=int(input("enter the element"))
    l1.append(x)
prime_of_l1=[i for i in l1 if check_prime(i)]
print(f"The given list is",l1)
print(f"prime numbers of given list={prime_of_l1}")
```

OUTPUT:



LAB-CYCLE-2

6. Write a program to generate Fibonacci series using loops.
7. Sort strings by length using lambda function.
8. Sort dictionary using multiple values and lambda function.

AIM:6. Write a program to generate Fibonacci series using loops.

PROGRAM:

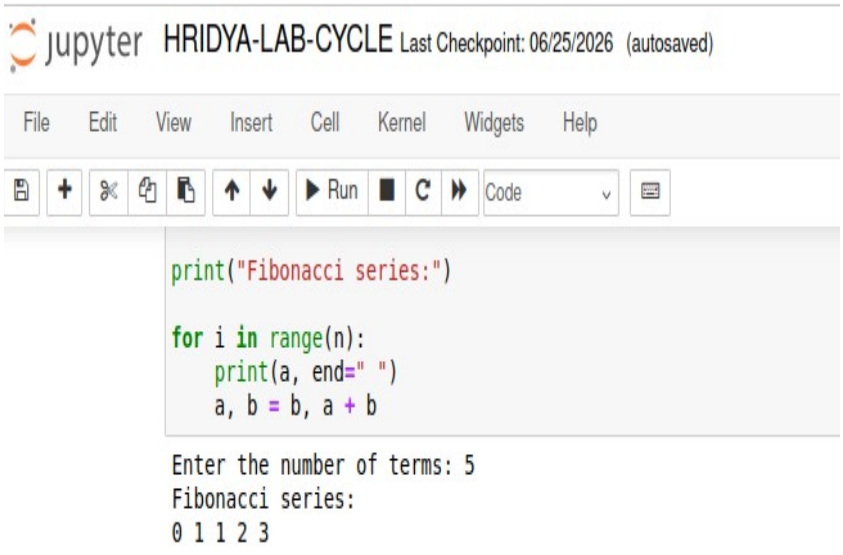
```
n = int(input("Enter the number of terms: "))

a = 0
b = 1

print("Fibonacci series:")

for i in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

OUTPUT:



LAB CYCLE 3

9.a)write a program to demonstrate polymorphism using function overloading and operator overloading.
b)write a program to demonstrate Inheritance and polymorphism.
10. Write a program to read and write text files .Use exceptions.

AIM:9.a)Write a program to demonstrate polymorphism using function overloading and operator overloading.

PROGRAM:

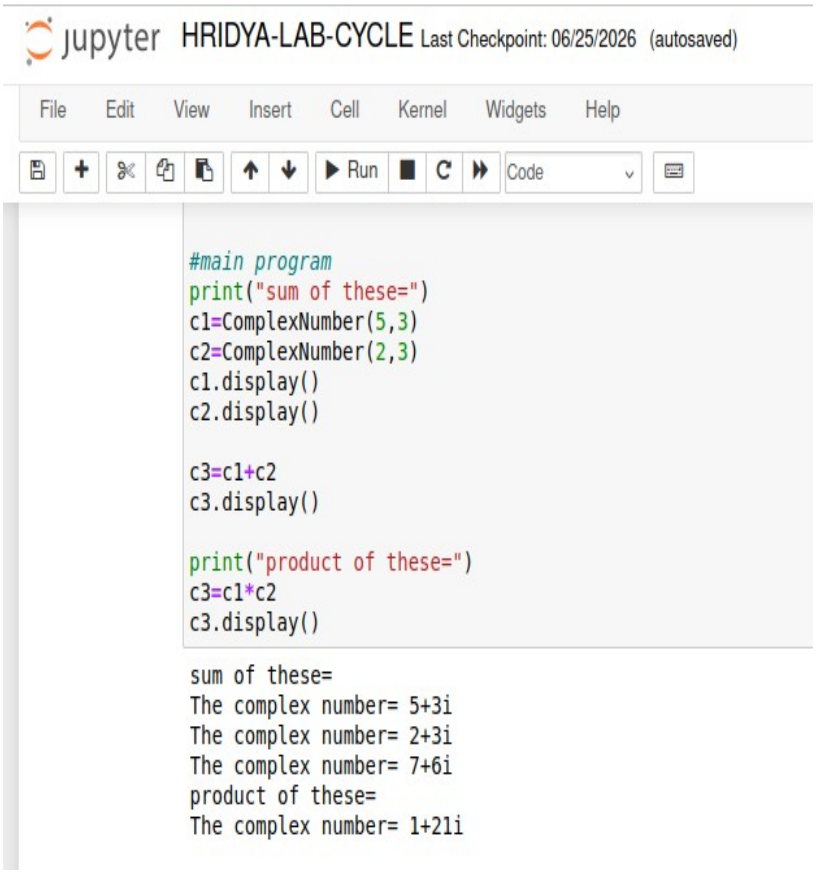
```
class ComplexNumber:
    real=0
    img=0
    def __init__(self,rel,ig):
        self.real=rel
        self.img=ig
    def display(self):
        print(f"The complex number= {self.real}+{self.img}i")
    def __add__(self,other):
        real=self.real+other.real
        img=self.img+other.img
        c3=ComplexNumber(real,img)
        return c3
         #return complex number(real,img)
    def __mul__(self,other):
        real=self.real*other.real-self.img*other.img
        img=self.real*other.img+self.img*other.real
        return ComplexNumber(real,img)


#main program
print("sum of these=")
c1=ComplexNumber(5,3)
c2=ComplexNumber(2,3)
c1.display()
c2.display()

c3=c1+c2
c3.display()

print("product of these=")
c3=c1*c2
c3.display()
```

OUTPUT:


jupyter HRIDYA-LAB-CYCLE Last Checkpoint: 06/25/2026 (autosaved)
File Edit View Insert Cell Kernel Widgets Help
Run
Code
#main program
print("sum of these=")
c1=ComplexNumber(5,3)
c2=ComplexNumber(2,3)
c1.display()
c2.display()
c3=c1+c2
c3.display()
print("product of these=")
c3=c1*c2
c3.display()
sum of these=
The complex number= 5+3i
The complex number= 2+3i
The complex number= 7+6i
product of these=
The complex number= 1+21i

AIM:b)write a program to demonstrate Inheritance and polymorphism.

PROGRAM:

```
class Person:
    def __init__(self, name, department):
        self.name = name
        self.department = department

    def display(self):
        print("NAME:", self.name)
        print("DEPARTMENT:", self.department)


class Student(Person):
    def __init__(self, name, department, perc):
        super().__init__(name, department)
        self.perc = perc

    def findgrade(self):
        if self.perc >= 90:
            print("Grade: A")
        elif self.perc >= 80:
            print("Grade: B")
        elif self.perc >= 70:
            print("Grade: C")
        else:
            print("Grade: D")

    def display(self):
        print("NAME:", self.name)
        print("DEPARTMENT:", self.department)
        print("PERCENTAGE:", self.perc)


class Teacher(Person):
    def __init__(self, name, department, salary):
        super().__init__(name, department)
        self.salary = salary

    def teach(self):
        print(self.name, "has salary")
        print(self.salary)


s1 = Student("HRIDYA", "BCA", 97)
```

```
s2 = Student("ANU", "BBA", 57)

t1 = Teacher("RADHU", "BBA", 130000)
t2 = Teacher("DEVU", "BCA", 100000)

s1.findgrade()
s1.display()

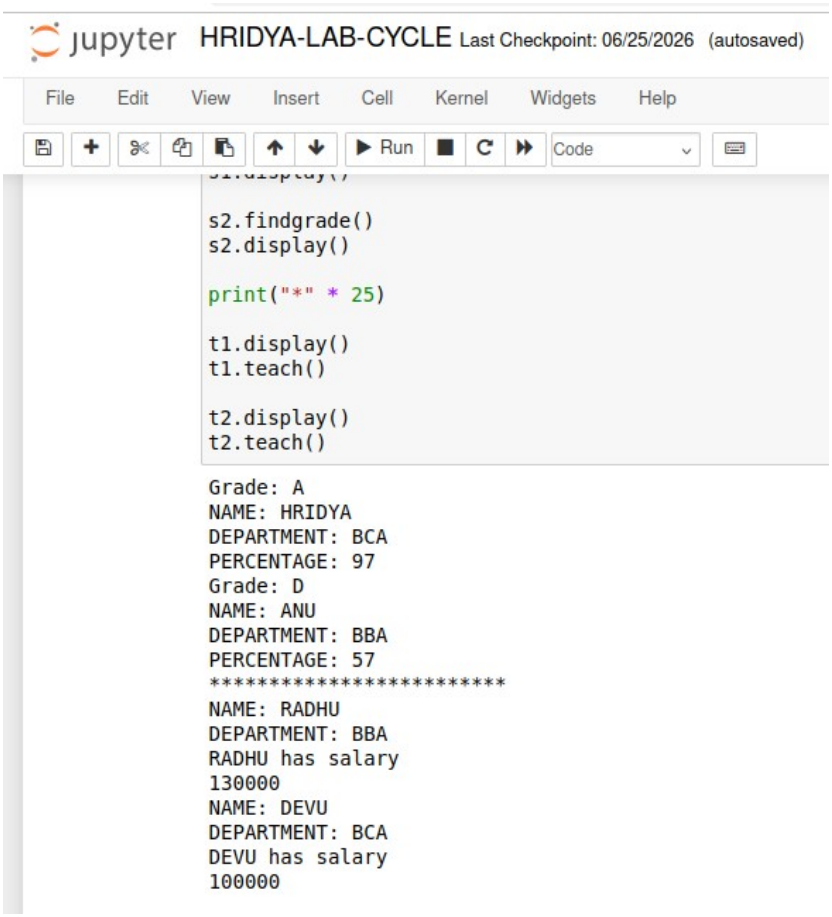
s2.findgrade()
s2.display()

print("*" * 25)

t1.display()
t1.teach()

t2.display()
t2.teach()
```

OUTPUT:



LAB CYCLE 4

11. Write a Python program using Pandas
a.to create and analyze a student dataset containing student name, marks and grade.
b. Use the following data: data = { "Name": ["Aman", "Diya" " Rahul", "Sneha","Arjun"], "Marks": [78, 92, 85, 67, 74], "Grade": ["B", "A", "A", "C", "B"]}
c. Perform the following operations:
i. Create a Pandas DataFrame using the given dataset.
ii. Display the complete dataset.
iii. Display the first 3 rows of the dataset.
iv. Check whether the dataset contains missing values.
v. Create a histogram showing mark distribution.
vi. Find median and standard deviation of marks.

12. Create dataframe from the given:
a. data = {"Name": ["Aman", "Diya", "Rahul", "Sneha", Arjun"], "Department": ["CS", "IT", "CS", "ECE", "IT"], "Marks": [78, 92, 85, 67, 74], "Attendance": [85, 90, 88, 75, 80], "Grade": ["B", "A", "A", "C", "B"]}.
b. Display Dataset Information
c. Display Statistical Summary
d. Display only student names and marks
e. Display students scoring more than 80 marks.
f. Sort students based on marks in descending order.
g. Add a new column named “Bonus” containing: 5 marks bonus for every student.

13. Using the above dataframe answer the following questions.
a. Create a new column named “Total” by adding marks and bonus.
b. Find average marks department-wise.
c. Count the number of students in each grade category.
d. Calculate: mean, median,standard deviation of student marks.
e. Display students having maximum attendance.
f. Increase marks of all students by 5.

14. Write a Python program using Pandas to clean and preprocess the following student dataset.
a. Perform the following operations:
i. Create a DataFrame containing student details.
ii. Identify missing values in the dataset.

iii. Replace missing values in the Marks column using the mean value.
iv. Remove duplicate rows from the dataset.
v. Convert student names into uppercase.
vi. Rename the column Marks to Score.
vii. Display the cleaned dataset.

AIM:11. Write a Python program using Pandas:

a.to create and analyze a student dataset containing student name, marks and grade.

b. Use the following data: data = { "Name": ["Aman", "Diya"
" Rahul", "Sneha","Arjun"], "Marks": [78, 92, 85, 67, 74], "Grade": ["B", "A", "A", "C", "B"]}

c. Perform the following operations:

i. Create a Pandas DataFrame using the given dataset.

ii. Display the complete dataset.

iii. Display the first 3 rows of the dataset.

iv. Check whether the dataset contains missing values.

v. Create a histogram showing mark distribution.

vi. Find median and standard deviation of marks. .

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {"Name": ["Aman", "Diya", "Rahul", "Sneha", "Arjun"],
       "Marks": [78, 92, 85, 67, 74],
       "Grade": ["B", "A", "A", "C", "B"]}

student = pd.DataFrame(data)

print("COMPLETE DATASET:")
print(student)

print("FIRST 3 ROWS:")
print(student.head(3))

print("MISSING VALUES:")
print(student.isnull().sum())

plt.figure(figsize=(5, 3))
plt.hist(student["Marks"],
        bins=7,
        color="pink",
        edgecolor="black")

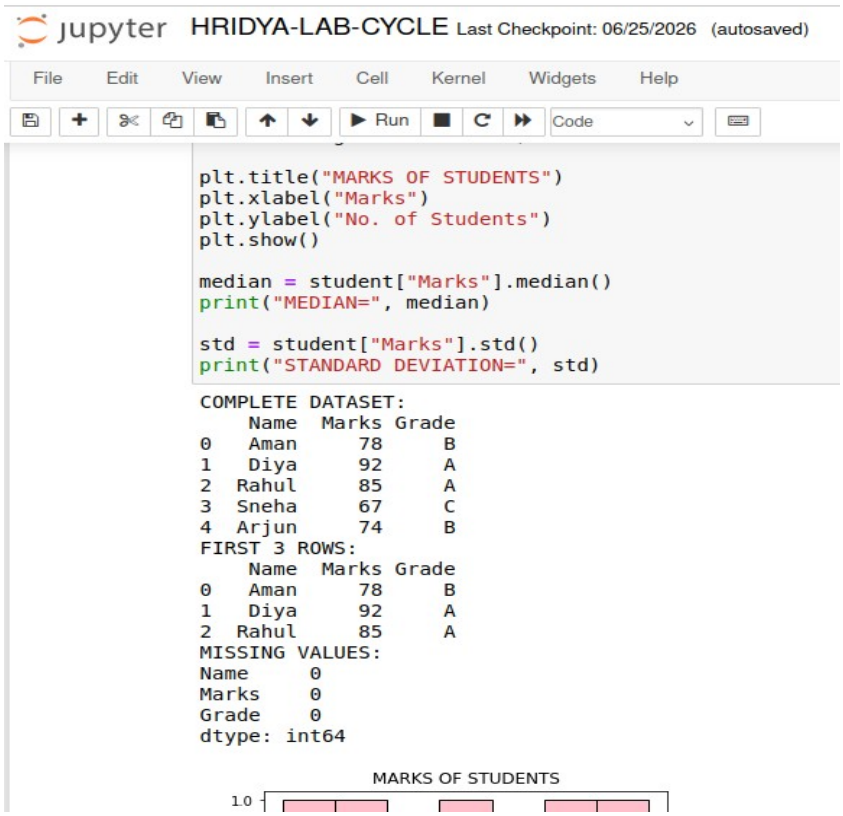
plt.title("MARKS OF STUDENTS")
plt.xlabel("Marks")
plt.ylabel("No. of Students")
plt.show()

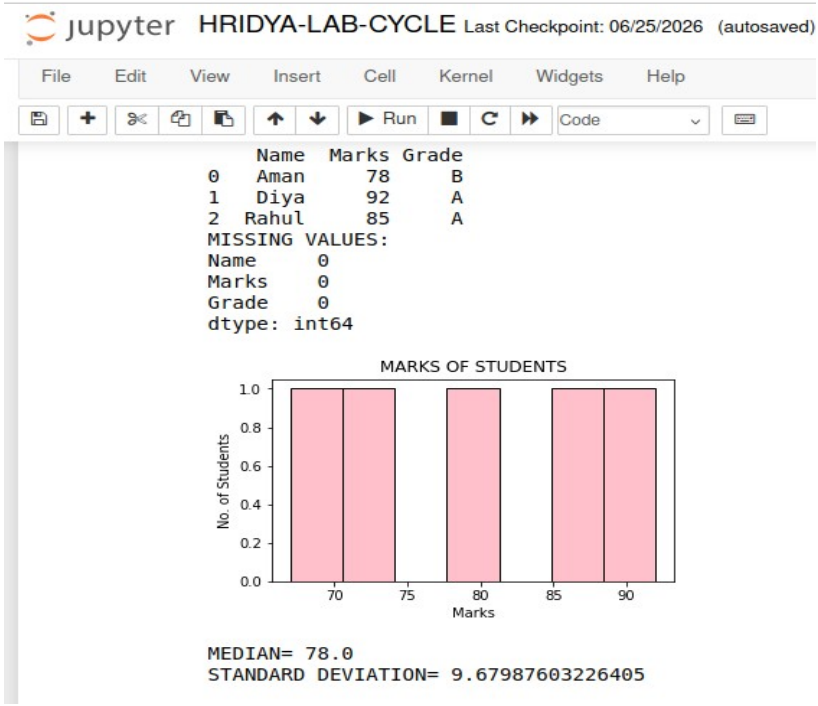
median = student["Marks"].median()
```

```
print("MEDIAN=", median)

std = student["Marks"].std()
print("STANDARD DEVIATION=", std)
```

OUTPUT:





AIM:12. Create dataframe from the given:

a. data = {"Name": ["Aman", "Diya", "Rahul", "Sneha", Arjun"], "Department":
["CS", "IT", "CS", "ECE", "IT"], "Marks": [78, 92, 85, 67, 74], "Attendance":
[85, 90, 88, 75, 80], "Grade": ["B", "A", "A", "C", "B"]}.

b. Display Dataset Information

c. Display Statistical Summary

d. Display only student names and marks

e. Display students scoring more than 80 marks.

f. Sort students based on marks in descending order.

g. Add a new column named “Bonus” containing: 5 marks bonus for every student.

PROGRAM:

```
import pandas as pd

data = {
   "Name": ["aman", "diya", "rahul", "sneha", "arjun"],
   "Department": ["CS", "IT", "CS", "ECE", "IT"],
   "Marks": [78, 92, 85, 67, 74],
   "Attendance": [85, 90, 88, 75, 80],
   "Grade": ["B", "A", "A", "C", "B"]
}

df = pd.DataFrame(data)

print("DataFrame Information:")
print(df.info())

print("\nStatistical Summary:")
print(df.describe())

print("\nStudent Name and Marks:")
print(df[["Name", "Marks"]])

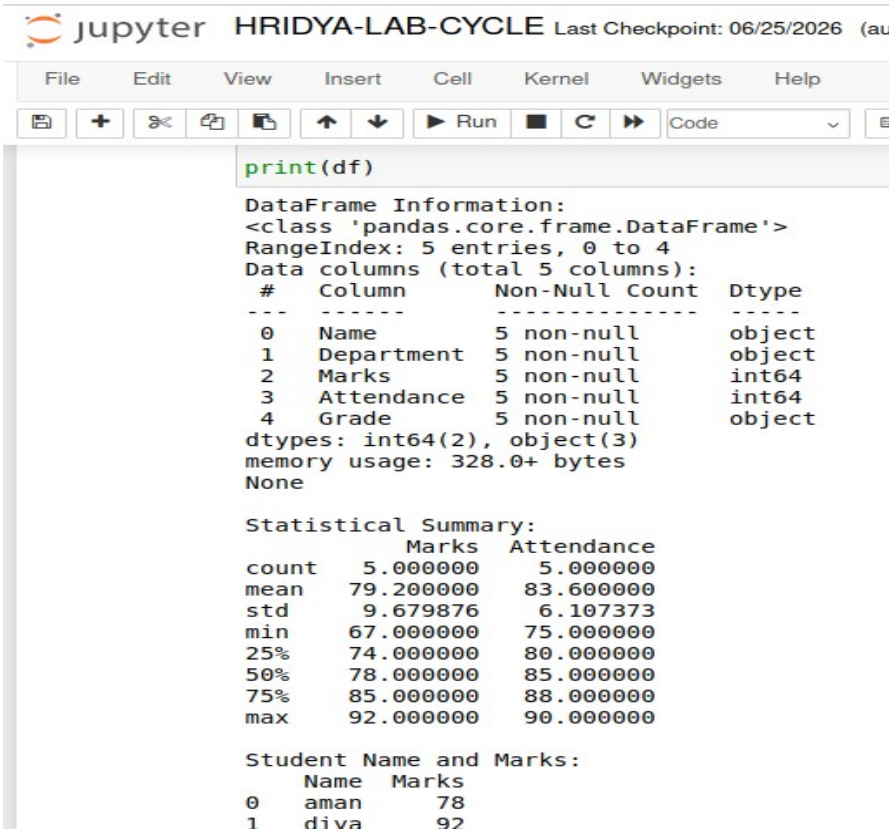
print("\nStudents scoring more than 80:")
print(df[df["Marks"] > 80])

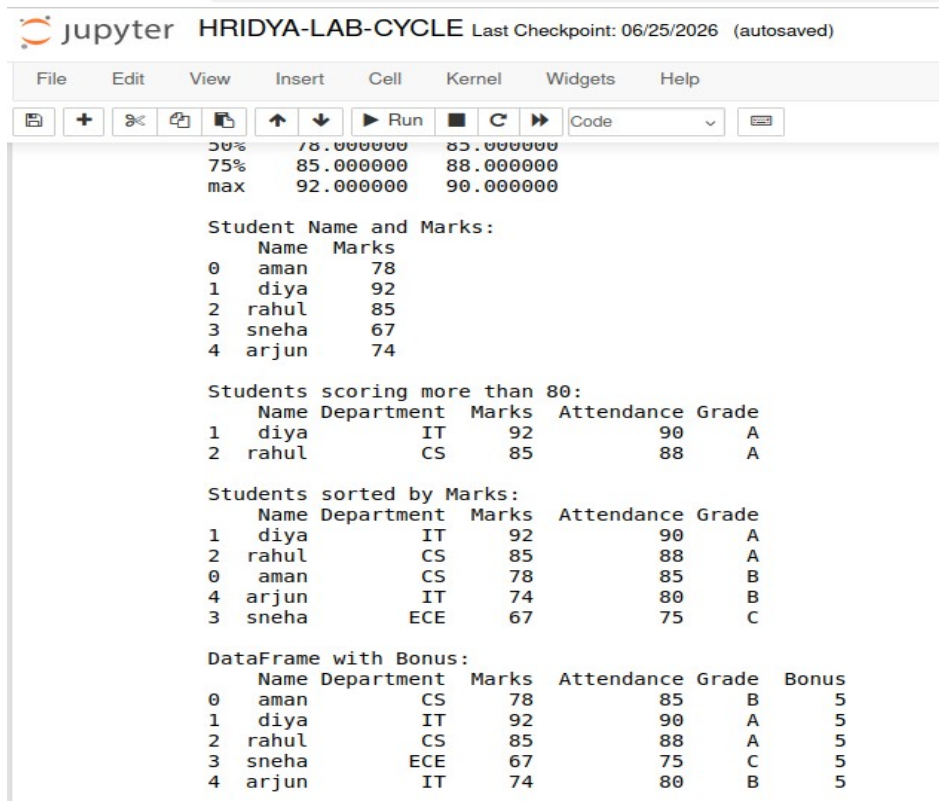
print("\nStudents sorted by Marks:")
print(df.sort_values(by="Marks", ascending=False))

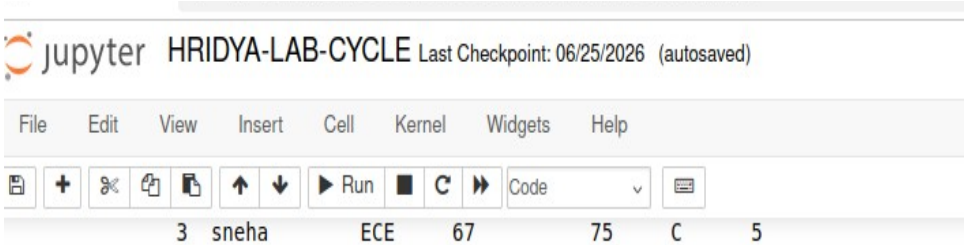
df["Bonus"] = 5

print("\nDataFrame with Bonus:")
print(df)
```

OUTPUT:


jupyter HRIDYA-LAB-CYCLE Last Checkpoint: 06/25/2026 (au
File Edit View Insert Cell Kernel Widgets Help
Run Code
print(df)
DataFrame Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
Column Non-Null Count Dtype
0 Name 5 non-null object
1 Department 5 non-null object
2 Marks 5 non-null int64
3 Attendance 5 non-null int64
4 Grade 5 non-null object
dtypes: int64(2), object(3)
memory usage: 328.0+ bytes
None
Statistical Summary:
Marks Attendance
count 5.000000 5.000000
mean 79.200000 83.600000
std 9.679876 6.107373
min 67.000000 75.000000
25% 74.000000 80.000000
50% 78.000000 85.000000
75% 85.000000 88.000000
max 92.000000 90.000000
Student Name and Marks:
Name Marks
0 aman 78


jupyter HRIDYA-LAB-CYCLE Last Checkpoint: 06/25/2026 (autosaved)
File Edit View Insert Cell Kernel Widgets Help
Run Code
50% 78.000000 85.000000
75% 85.000000 88.000000
max 92.000000 90.000000
Student Name and Marks:
Name Marks
0 aman 78
1 diya 92
2 rahul 85
3 sneha 67
4 arjun 74
Students scoring more than 80:
Name Department Marks Attendance Grade
1 diya IT 92 90 A
2 rahul CS 85 88 A
Students sorted by Marks:
Name Department Marks Attendance Grade
1 diya IT 92 90 A
2 rahul CS 85 88 A
0 aman CS 78 85 B
4 arjun IT 74 80 B
3 sneha ECE 67 75 C
DataFrame with Bonus:
Name Department Marks Attendance Grade Bonus
0 aman CS 78 85 B 5
1 diya IT 92 90 A 5
2 rahul CS 85 88 A 5
3 sneha ECE 67 75 C 5
4 arjun IT 74 80 B 5




```
3  sneha        ECE     67          75     C      5
4  arjun         IT     74          80     B      5

_____________________________________________
    Name Department  Marks  Attendance Grade  Bonus  Total
0   aman         CS     78          85     B      5     83
1   diya         IT     92          90     A      5     97
2  rahul         CS     85          88     A      5     90
3  sneha        ECE     67          75     C      5     72
4  arjun         IT     74          80     B      5     79
Department
CS     81.5
ECE    67.0
IT     83.0
Name: Marks, dtype: float64
B    2
A    2
C    1
Name: Grade, dtype: int64
Mean: 79.2
Median: 78.0
Standard Deviation: 9.67987603226405
   Name Department  Marks  Attendance Grade  Bonus  Total
1  diya         IT     92          90     A      5     97
    Name Department  Marks  Attendance Grade  Bonus  Total
0   aman         CS     83          85     B      5     83
1   diya         IT     97          90     A      5     97
2  rahul         CS     90          88     A      5     90
3  sneha        ECE     72          75     C      5     72
4  arjun         IT     79          80     B      5     79
```

AIM:13. Using the above dataframe answer the following questions.
a. Create a new column named “Total” by adding marks and bonus.
b. Find average marks department-wise.
c. Count the number of students in each grade category.
d. Calculate: mean, median,standard deviation of student marks.
e. Display students having maximum attendance.
f. Increase marks of all students by 5.

PROGRAM:

```
df["Total"] = df["Marks"] + df["Bonus"]
print(df)

print(df.groupby("Department")["Marks"].mean())

print(df["Grade"].value_counts())

print("Mean:", df["Marks"].mean())

print("Median:", df["Marks"].median())

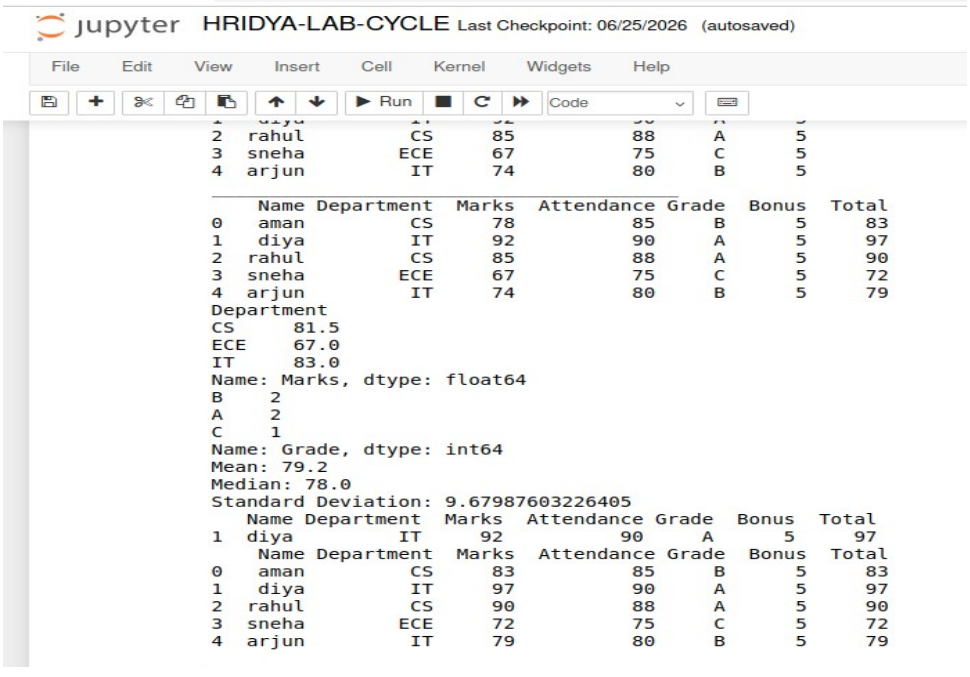
print("Standard Deviation:", df["Marks"].std())

print(df[df["Attendance"] == df["Attendance"].max()])

df["Marks"] = df["Marks"] + 5

print(df)
```

OUTPUT:



LAB CYCLE 5

15. Plot simple mathematical functions.[y=3x^2+4x]
16. Create simple login form using streamlit
17. Create student registration form using streamlit
18. Sample program to use Git and Github

AIM:15. Plot simple mathematical functions.[y=3x^2+4x]

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-10, 10, 100)
y = 2*x**2 + 3*x + 1

plt.plot(x, y)
plt.xlabel("x")
plt.ylabel("y")
plt.title("y = 2x² + 3x + 1")
plt.grid()
plt.show()
```

OUPUT:

